

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$\frac{d}{dx}(3y^2) = 6y \frac{dy}{dx}$ or $\frac{d}{dx}(qxy) = qx \frac{dy}{dx} + qy$	M1	2.1
	$3px^2 + qx \frac{dy}{dx} + qy + 6y \frac{dy}{dx} = 0$	A1	1.1b
	$(qx + 6y) \frac{dy}{dx} = -3px^2 - qy \Rightarrow \frac{dy}{dx} = \dots$	dM1	2.1
	$\frac{dy}{dx} = \frac{-3px^2 - qy}{qx + 6y}$	A1	1.1b
		(4)	
(b)	$p(-1)^3 + q(-1)(-4) + 3(-4)^2 = 26$	M1	1.1b
	$19x + 26y + 123 = 0 \Rightarrow m = -\frac{19}{26}$	B1	2.2a
	$\frac{-3p(-1)^2 - q(-4)}{q(-1) + 6(-4)} = \frac{26}{19} \quad \text{or} \quad \frac{q(-1) + 6(-4)}{3p(-1)^2 + q(-4)} = -\frac{19}{26}$	M1	3.1a
	$p - 4q = 22, \quad 57p - 102q = 624 \Rightarrow p = \dots, q = \dots$	dM1	1.1b
	$p = 2, \quad q = -5$	A1	1.1b
		(5)	
(9 marks)			

Notes

(a)

M1: For selecting the appropriate method of differentiating:

Allow this mark for either $3y^2 \rightarrow \alpha y \frac{dy}{dx}$ or $qxy \rightarrow \alpha x \frac{dy}{dx} + \beta y$

A1: Fully correct differentiation. Ignore any spurious $\frac{dy}{dx} = \dots$

dM1: A valid attempt to make $\frac{dy}{dx}$ the subject with 2 terms only in $\frac{dy}{dx}$ coming from qxy and $3y^2$

Depends on the first method mark.

A1: Fully correct expression

(b)

M1: Uses $x = -1$ and $y = -4$ in the equation of C to obtain an equation in p and q

B1: Deduces the correct gradient of the given normal.

This may be implied by e.g.

$$19x + 26y + 123 = 0 \Rightarrow y = -\frac{19}{26}x + \dots \Rightarrow \text{Tangent equation is } y = \frac{26}{19}x + \dots$$

M1: Fully correct strategy to establish an equation connecting p and q using $x = -1$ and $y = -4$ in their $\frac{dy}{dx}$ and the gradient of the normal. E.g. $(a) = -1 \div \text{their } -\frac{19}{26}$ or $-1 \div (a) = \text{their } -\frac{19}{26}$

dM1: Solves simultaneously to obtain values for p and q .

Depends on both previous method marks.

A1: Correct values

Note that in (b), attempts to form the equation of the normal in terms of p and q and then compare coefficients with $19x + 26y + 123 = 0$ score no marks. If there is any doubt use Review.

Q2.

Question	Scheme	Marks	AOs
(a)	$x^2 \tan y = 9 \Rightarrow 2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$	M1 A1	3.1a 1.1b
	Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$	M1	1.1b
	$\frac{dy}{dx} = \frac{-2x \times \frac{9}{x^2}}{x^2 \left(1 + \frac{81}{x^4}\right)} = \frac{-18x}{x^4 + 81} *$	A1*	2.1
		(4)	
(b)	$\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$ $\Rightarrow \frac{d^2y}{dx^2} = \frac{-18 \times (x^4 + 81) - (-18x)(4x^3)}{(x^4 + 81)^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2}$ o.e.	M1 A1	1.1b 1.1b
	States that when $x < \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} < 0$ when $x = \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} = 0$ AND when $x > \sqrt[4]{27} \Rightarrow \frac{d^2y}{dx^2} > 0$ giving a point of inflection when $x = \sqrt[4]{27}$	A1	2.4
		(3)	
		(7 marks)	
Notes:			

(a)

M1: Attempts to differentiate $\tan y$ implicitly. Eg. $\tan y \rightarrow \sec^2 y \frac{dy}{dx}$ or $\cot y \rightarrow -\operatorname{cosec}^2 y \frac{dy}{dx}$

You may well see an attempt $\tan y = \frac{9}{x^2} \Rightarrow \sec^2 y \frac{dy}{dx} = \dots$

When a candidate writes $x^2 \tan y = 9 \Rightarrow x = 3 \tan^{\frac{1}{2}} y$ the mark is scored for $\tan^{\frac{1}{2}} y \rightarrow \dots \tan^{\frac{3}{2}} y \sec^2 y$

A1: Correct differentiation $2x \tan y + x^2 \sec^2 y \frac{dy}{dx} = 0$

Allow also $\sec^2 y \frac{dy}{dx} = -\frac{18}{x^3}$ or $2x = -9 \operatorname{cosec}^2 y \frac{dy}{dx}$ amongst others

M1: Full method to get $\frac{dy}{dx}$ in terms of x using $\sec^2 y = 1 + \tan^2 y = 1 + f(x)$

A1*: Proceeds correctly to the given answer of $\frac{dy}{dx} = \frac{-18x}{x^4 + 81}$

(b)

M1: Attempts to differentiate the given expression using the product or quotient rule.

For example look for a correct attempt at $\frac{vu' - uv'}{v^2}$ with $u = -18x, v = x^4 + 81, u' = \pm 18, v' = \dots x^3$

If no method is seen or implied award for $\frac{\pm 18 \times (x^4 + 81) \pm 18x(ax^3)}{(x^4 + 81)^2}$

Using the product rule award for $\pm 18(x^4 + 81)^{-1} \pm 18x(x^4 + 81)^{-2} \times cx^3$

A1: Correct simplified $\frac{d^2y}{dx^2} = \frac{54(x^4 - 27)}{(x^4 + 81)^2}$ o.e. such as $\frac{d^2y}{dx^2} = \frac{54x^4 - 1458}{(x^4 + 81)^2}$

Alternatively score for showing that when a correct (unsimplified) $\frac{d^2y}{dx^2} = 0 \Rightarrow x^4 = 27 \Rightarrow x = \sqrt[4]{27}$

Or for substituting $x = \sqrt[4]{27}$ into an unsimplified but correct $\frac{d^2y}{dx^2}$ and showing that it is 0

A1: Correct explanation with a minimal conclusion and correct second derivative.

See scheme.

It can be also be argued from $x^4 < 27, x^4 = 27$ and $x^4 > 27$ provided the conclusion states that the point of inflection is at $x = \sqrt[4]{27}$

Alternatively substitutes values of x either side of $\sqrt[4]{27}$ and at $\sqrt[4]{27}$, into $\frac{d^2y}{dx^2}$, finds all three values and makes a minimal conclusion.

A different method involves finding $\frac{d^3y}{dx^3}$ and showing that $\frac{d^3y}{dx^3} \neq 0$ and $\frac{d^2y}{dx^2} = 0$ when $x = \sqrt[4]{27}$

$$\text{FYI } \frac{d^3y}{dx^3} = \frac{23328x^3}{(x^4 + 81)^3} = 0.219 \text{ when } x = \sqrt[4]{27}$$

Alternative part (a) using arctan

M1: Sets $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times \dots$ where ... could be 1

A2: $y = \arctan \frac{9}{x^2} \rightarrow \frac{dy}{dx} = \frac{1}{1 + \left(\frac{9}{x^2}\right)^2} \times -\frac{18}{x^3}$

A1*: $\frac{dy}{dx} = \frac{1}{1 + \frac{81}{x^4}} \times -\frac{18}{x^3} = \frac{-18x}{x^4 + 1}$ showing correct intermediate step and no errors.

Q3.

Question	Scheme	Marks	AOs
(a)	Either $3y^2 \rightarrow Ay \frac{dy}{dx}$ or $2xy \rightarrow 2x \frac{dy}{dx} + 2y$	M1	2.1
	$2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$	A1	1.1b
	$(6y - 2x) \frac{dy}{dx} = 2y - 2x$	M1	2.1
	$\frac{dy}{dx} = \frac{2y - 2x}{6y - 2x} = \frac{y - x}{3y - x} *$	A1*	1.1b
	(4)		
(b)	(At P and Q $\frac{dy}{dx} \rightarrow \infty \Rightarrow$) Deduces that $3y - x = 0$	M1	2.2a
	Solves $y = \frac{1}{3}x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously	M1	3.1a
	$\Rightarrow x = (\pm)5\sqrt{3}$ OR $\Rightarrow y = (\pm)\frac{5}{3}\sqrt{3}$	A1	1.1b
	Using $y = \frac{1}{3}x \Rightarrow x = ..$ AND $y = ..$	dM1	1.1b
	$P = \left(-5\sqrt{3}, -\frac{5}{3}\sqrt{3}\right)$	A1	2.2a
	(5)		
(c)	Explains that you need to solve $y = x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously and choose the positive solution	B1ft	2.4
	(1)		
(10 marks)			

Notes:

(a)

M1: For selecting the appropriate method of differentiating either $3y^2 \rightarrow Ay \frac{dy}{dx}$ or $2xy \rightarrow 2x \frac{dy}{dx} + 2y$

It may be quite difficult awarding it for the product rule but condone $-2xy \rightarrow -2x \frac{dy}{dx} + 2y$ unless you see evidence that they have used the incorrect law $uv' - uv'$

A1: Fully correct derivative $2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$

Allow attempts where candidates write $2xdx - 2xdy - 2ydx + 6ydy = 0$

but watch for students who write $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx}$ This, on its own, is A0 unless you are

convinced that this is just their notation. Eg $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y \frac{dy}{dx} = 0$

M1: For a valid attempt at making $\frac{dy}{dx}$ the subject, with two terms in $\frac{dy}{dx}$ coming from $3y^2$ and $2xy$ Look for $(\dots \pm \dots) \frac{dy}{dx} = \dots$. It is implied by $\frac{dy}{dx} = \frac{2y-2x}{6y-2x}$ This cannot be scored from attempts such as $\frac{dy}{dx} = 2x - 2x \frac{dy}{dx} - 2y + 6y$ which only has one correct term. A1*: $\frac{dy}{dx} = \frac{y-x}{3y-x}$ with no errors or omissions. The previous line $\frac{dy}{dx} = \frac{2y-2x}{6y-2x}$ or equivalent must be seen.	
(b) M1: Deduces that $3y - x = 0$ oe M1: Attempts to find either the x or y coordinates of P and Q by solving their $y = \frac{1}{3}x$ with $x^2 - 2xy + 3y^2 = 50$ simultaneously. Allow for finding a quadratic equation in x or y and solving to find at least one value for x or y. This may be awarded when candidates make the numerator = 0 ie using $y = x$ A1: $\Rightarrow x = (\pm)5\sqrt{3}$ OR $\Rightarrow y = (\pm)\frac{5}{3}\sqrt{3}$ dM1: Dependent upon the previous M, it is for finding the y coordinate from their x (or vice versa) This may also be scored following the numerator being set to 0 ie using $y = x$ A1: Deduces that $P = \left(-5\sqrt{3}, -\frac{5}{3}\sqrt{3}\right)$ OE. Allow to be $x = \dots$ $y = \dots$	
(c) Blft: Explains that this is where $\frac{dy}{dx} = 0$ and so you need to solve $y = x$ and $x^2 - 2xy + 3y^2 = 50$ simultaneously and choose the positive solution (or larger solution). Allow a follow through for candidates who mix up parts (b) and (c) Alternatively candidates could complete the square $(x-y)^2 + 2y^2 = 50$ and state that y would reach a maximum value when $x = y$ and choose the positive solution from $2y^2 = 50$	